

CTS SRE

Fault Vulnerability Analysis 1& 2

EMPOWERING High Performance
Technology Teams

Agenda for the Day

- Introduction to Fault Vulnerability Analysis**
- Analyzing Faults in Application, Database, and Infrastructure Layers**
- Developing Resilience Strategies**
- Distributed System Vulnerabilities & External Dependencies**
- GenAI-Powered Fault Analysis**
- Proactive Fault Recovery Mechanisms**
- Real-World Case Studies & Critical Outage Analysis**
- Summary & Key Takeaways**

Introduction to Fault Vulnerability Analysis

What is Fault Vulnerability Analysis?

A structured approach to identifying, assessing, and mitigating risks in distributed systems.

Why is it important?

Avoid unexpected downtime, increase resilience, and ensure business continuity.

Key Focus Areas:

Database, Application, and Infrastructure Layers

Third-party Dependencies

AI-driven anomaly detection

Fault Analysis in Application, Database, and Infrastructure Layers

Application Layer Faults:

Unhandled exceptions, **memory leaks**, **thread deadlocks**.

High CPU utilization due to inefficient code.

Security misconfigurations leading to vulnerabilities.

Database Layer Faults:

Slow queries, deadlocks, and replication lag.

Single point of failure in database architecture.

Inconsistent backups and lack of disaster recovery plans.

Infrastructure Layer Faults:

Cloud resource failures (EC2 crashes, disk failures, network outages).

Auto-scaling misconfigurations leading to under/over provisioning.

Dependency failures (DNS resolution issues, IAM permission errors).

 **Example: A high-traffic e-commerce application suffering from database contention issues during Black Friday sales.**

Developing Resilience Strategies

◆ Redundancy Mechanisms:

Multi-region deployments with **active-active** configurations.

Data replication across **multiple availability** zones.

◆ Failover Protocols:

Graceful degradation strategies to keep critical services running.

Load balancing across multiple data centers.

◆ Self-Healing Systems:

Automated rollback strategies with **blue-green & canary deployments**.

Circuit breakers & rate limiters to handle traffic surges.

🔗 **Example:** How **Netflix Chaos Monkey** proactively tests system resilience.

Distributed System Vulnerabilities & External Dependencies

Challenges of Distributed Architectures:

Cascading failures due to interdependent services.

Handling network partitions and data consistency issues.

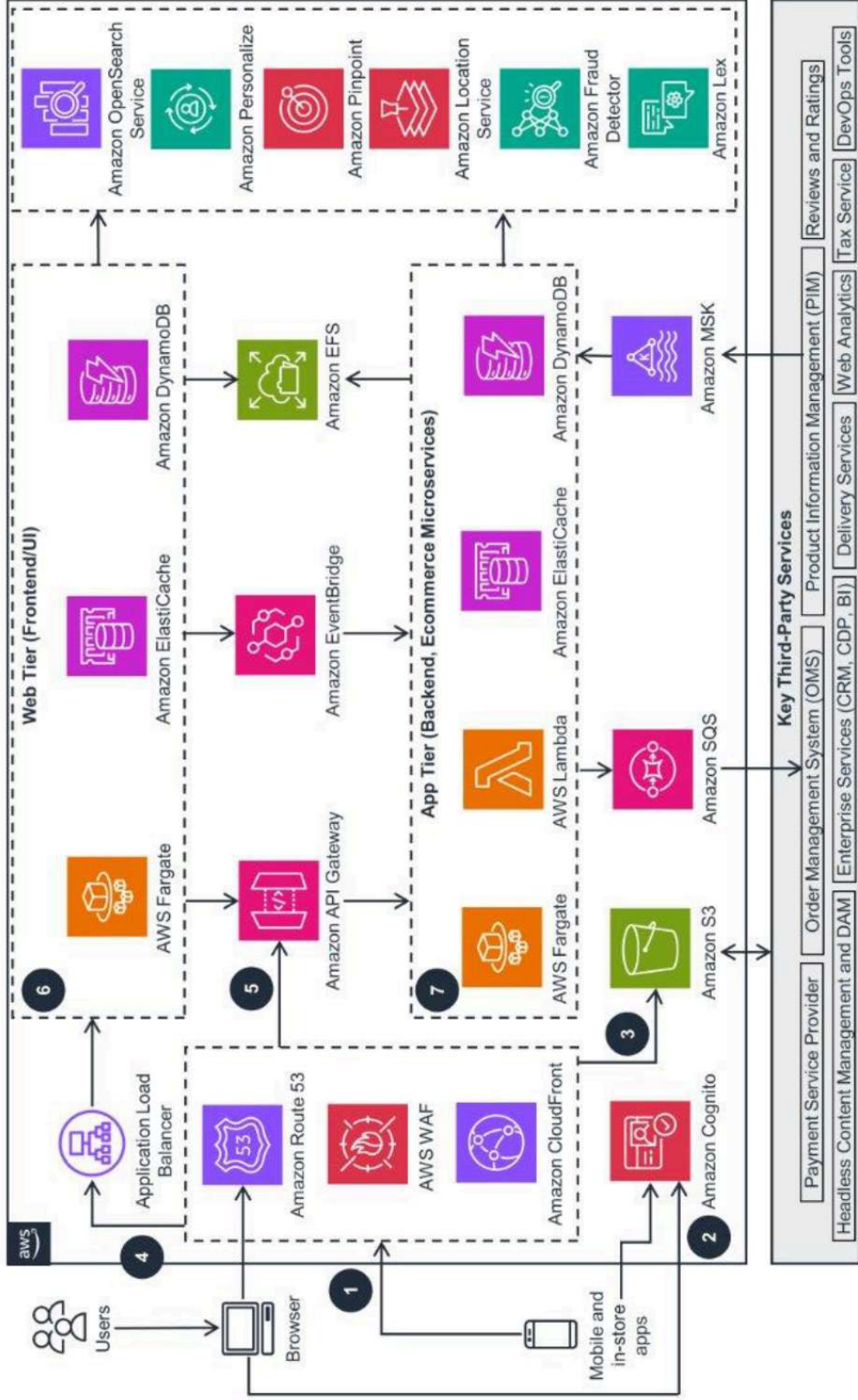
External Dependencies & Third-Party APIs:

API rate limits and throttling causing downtime.

Dependency on external CDNs, payment gateways, and messaging services.

Lack of **fallback mechanisms** when external services fail.

 **Example: The AWS S3 Outage (2017)** that caused failures in multiple dependent services.



GenAI-Powered Fault Analysis



Leveraging GenAI for Automated Fault Analysis:

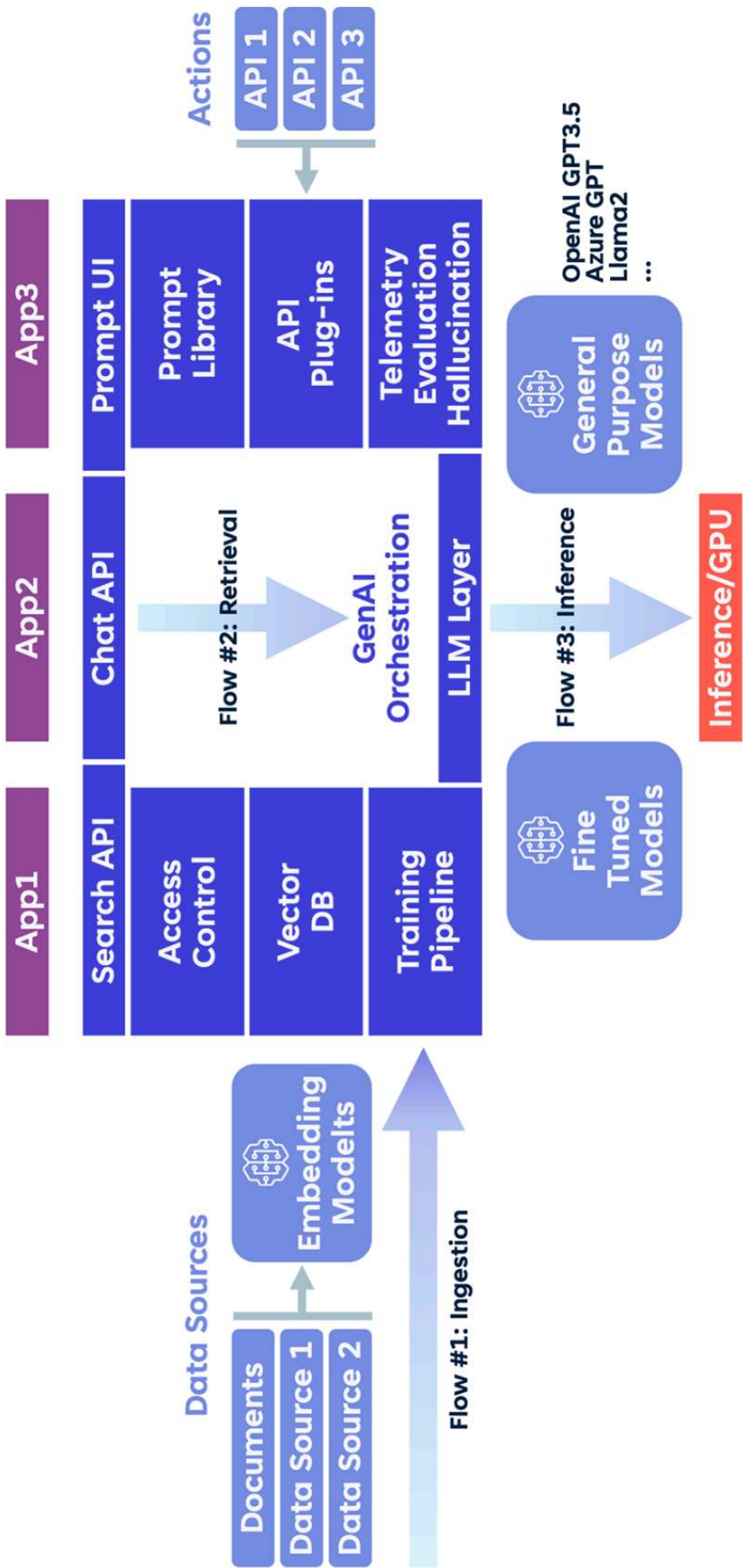
- Analyzing system logs, metrics, and traces to detect anomalies.
- Pattern recognition in historical incidents to predict failures.
- Intelligent root cause analysis (RCA) reducing manual intervention.

◆ AI-Powered Anomaly Detection:

- Identifying abnormal patterns in application behavior.
- Detecting slow database queries **before they cause bottlenecks.**
- Analyzing cloud infrastructure telemetry **to predict failures.**

 **Example: AI-driven fault prediction reducing incident response times by 40%.**

Reference Architecture for GenAI



Proactive Fault Recovery Mechanisms

✓ Database Resilience Strategies:

- Automated backup & recovery workflows.
- Read replicas & caching layers for load distribution.

✓ Self-Healing Kubernetes Clusters:

- Pod auto-restart policies for failing containers.
- Horizontal & vertical pod autoscaling.

✓ Automated Incident Response:

- Event-driven remediation using **AWS Lambda, PagerDuty, and Ansible.**
- **Self-recovery workflows triggered by GenAI insights.**

🔗 **Example: Facebook's self-healing database clusters preventing large-scale outages.**

Breakout

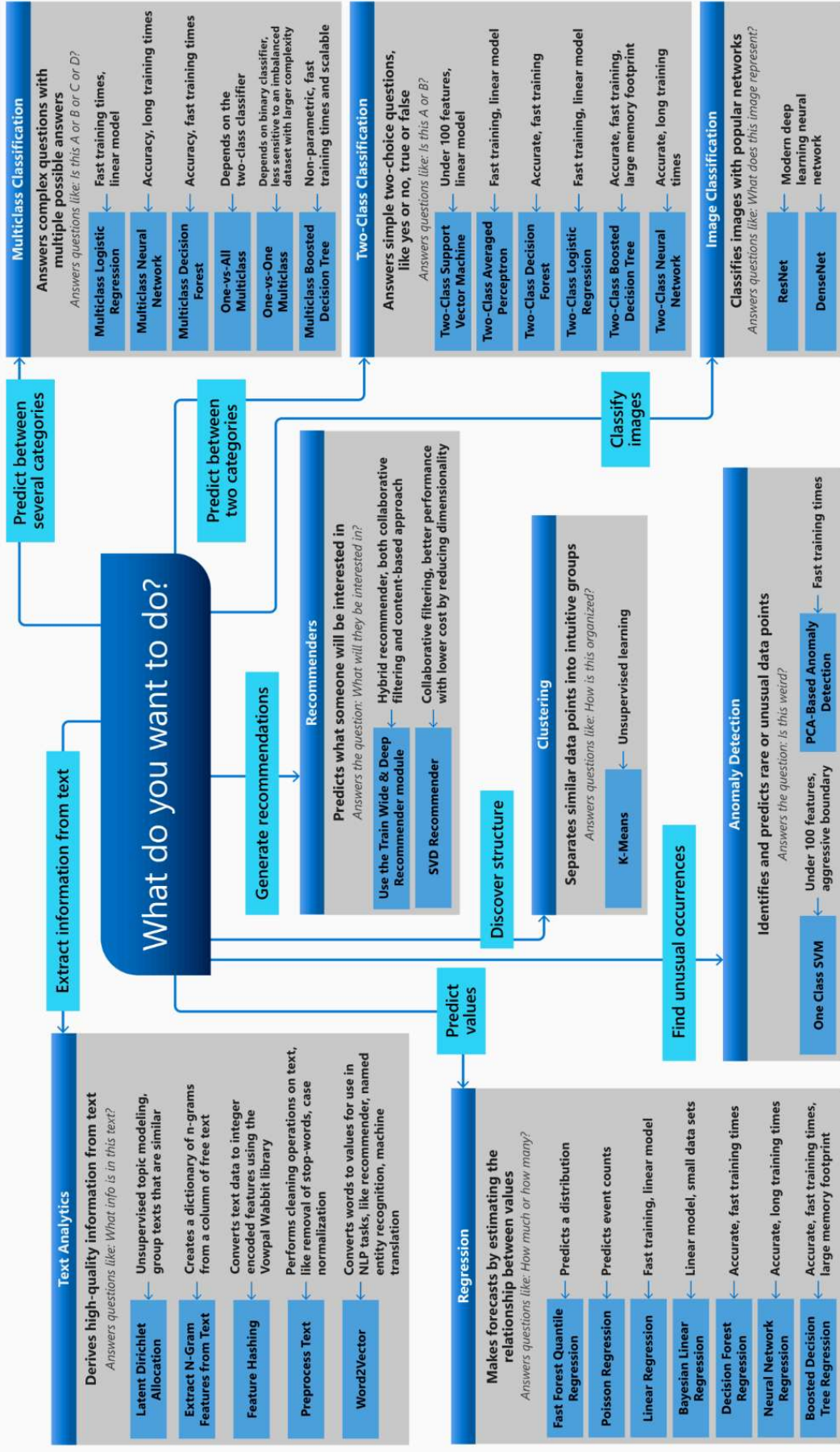
Summary & Key Takeaways

- ✓ **Proactive fault vulnerability analysis ensures system reliability.**
- ✓ **Resilience strategies reduce downtime and improve failover handling.**
- ✓ **GenAI enhances fault detection, anomaly prediction, and RCA.**
- ✓ **Chaos engineering validates and strengthens failure recovery strategies.**
- ✓ **A combination of redundancy, self-healing systems, and automated monitoring is critical for modern SRE practices.**



Machine Learning Algorithm Cheat Sheet

This cheat sheet helps you choose the best machine learning algorithm for your predictive analytics solution. Your decision is driven by both the nature of your data and the goal you want to achieve with your data.



Introduction to case studies

Expectation from case study



Participants are expected to work on the deliverables:

- Define **SLI/SLO metrics** and create monitoring dashboards.
- Automate cloud infrastructure provisioning and deployments.
- Run **chaos experiments** and document resilience gaps.
- Conduct **root cause analysis (RCA)** and propose improvements.

Thank You!
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